



Proposed CEA Link — December 2026

Weekday Lates with weeks 13–17 kept exactly as written, the evening gap filled, and the fatigue factors driven as low as this design allows

Prepared 17 September 2026 · built for the December 2026 timetable and assessed by the workspace's own rule modules · figures on this page are computed, not typed

PROPOSAL

Weekday Lates 3

Weeks 13–17 protected, fatigue at its floor

CODE **WL3-24-F7**

FINGERPRINT **a6234195**

BUILT FROM duty table — · seed — · 24 lines · 4 cover weeks

LINEAGE Weekday Lates (WL-24-EXT · a52ec588) with weeks 13–17 held as written, the 08:30–17:00 turns outside them re-timed into the evening, and the rotation then searched fatigue-first. Compare with Weekday Lates 2 (WL2-24-R21 · 33f70893), which had no protected weeks and could also move week 16's three turns.

✓ **24** lines designed

✓ **35h 00m** a week — exactly the contract

✓ **0** hard-limit breaches

✓ **0** rests under 12h

⚠ **1** fatigue factor present (today: 5)

13 of 19

shift times are today's

6 times are new; the rest are turns people already work

6 days

longest run of worked days

today's link reaches 9; Chiltern's limit is 13

12 of 20

working weeks are one turn

same clock time all week, as 7 of today's 16 are

4 in 24

full weekends off

17% of the rotation, against 20% today (4 in 20)

4 cover weeks

lines 1, 7, 12, 17

gaps of 8, 6, 5, 5 around the wheel — inherited from Weekday Lates, not re-spread

2h 23m a week

typical week-to-week move

how far the working day shifts each week — today's link moves 4h 0m a week

What this is

A proposal for the CEA link once the December 2026 timetable arrives, for a team of 24 rather than 20. The brief was to keep the **feel** of the link people work today — a week is one turn, rest days come in pairs, four whole cover weeks spread round the rotation, and the same 06:20 / 08:00 / 14:00 / 15:15 clock times — while meeting the staffing shape agreed for December (four to open every day, three through to the close, four on a Saturday, five still on at 22:00, fourteen on a Saturday, ten on a Sunday) and every rule the link is assessed against: Chiltern's 13-day limit, twelve hours between duties, the exact 35-hour contracted week, and the ORR's good-practice fatigue factors.

It was not drawn by this app. It arrived as a Word table — a proposal from outside the workspace — and was read straight into the app's own shape, then judged by the same modules that judge a design the workspace generated itself: runDesignChecks, weeklyHours, assessHardLimits, assessFatigue, scoreOrder and calcHourlyCoverage. There is no search behind it and nothing to reproduce; what can be checked is every figure on these pages, each computed from the cells on page 6. Paste those cells into Links → Import and the workspace restates all of it.

What it is not. Not a recommendation and not a finished roster: nothing here has been through the roster office or a rep. The 24-line length, the four cover weeks and the December headcounts are owner-relayed figures with no document behind them (evidence class C), and the 13-day limit's policy citation is still outstanding — pages 5 and 6 say so where it applies. It clears every rule the tool can check, which is the floor for being worth discussing.

THE ROTATION

The 24-line link

Sunday to Saturday per line; everyone moves down one line each week and line 24 goes back to line 1. Hours and days at the right, cover beneath.

LINE	SUN	MON	TUE	WED	THU	FRI	SAT	MON-SAT hours	ALL WEEK hours	DAYS WORKED Mon-Sat
1	SP	SP	SP	SP	SP	SP	SP	35h 00m	35h 00m	—
2	13:30 22:00	13:30 22:00	RD	RD	13:30 22:00	13:30 22:00	14:45 23:55	34h 40m	43h 10m	4
3	14:30 23:25	14:00 22:30	RD	RD	RD	07:00 15:30	06:20 14:50	25h 30m	34h 25m	3
4	07:15 15:45	06:20 13:45	06:20 13:45	RD	RD	06:20 13:45	08:00 16:00	30h 15m	38h 45m	4
5	07:15 15:45	06:20 13:45	06:20 13:45	06:20 13:45	RD	RD	06:20 14:50	30h 45m	39h 15m	4
6	07:15 15:45	06:20 13:45	06:20 13:35	06:20 13:35	06:20 13:35	RD	RD	29h 10m	37h 40m	4
7	SP	SP	SP	SP	SP	SP	SP	35h 00m	35h 00m	—
8	RD	16:25 23:55	16:25 23:55	16:25 23:55	16:25 23:55	16:25 23:55	RD	37h 30m	37h 30m	5
9	RD	16:25 23:55	16:25 23:55	16:25 23:55	16:25 23:55	16:25 23:55	16:25 23:55	45h 00m	45h 00m	6
10	RD	16:25 23:55	16:25 23:55	16:25 23:55	16:25 23:55	16:25 23:55	RD	37h 30m	37h 30m	5
11	RD	14:00 22:30	14:00 22:30	14:00 22:30	14:00 22:30	14:00 22:30	RD	42h 30m	42h 30m	5
12	SP	SP	SP	SP	SP	SP	SP	35h 00m	35h 00m	—
13	10:30 19:00	12:30 21:00	RD	RD	06:20 14:20	06:20 13:35	06:20 14:20	31h 45m	40h 15m	4
14	RD	RD	08:30 16:30	08:30 16:30	13:30 22:00	13:00 21:00	14:45 23:55	41h 40m	41h 40m	5
15	14:30 23:25	RD	07:00 15:30	06:20 13:45	06:20 13:45	06:20 14:20	RD	31h 20m	40h 15m	4
16	RD	06:20 14:20	12:30 21:00	13:30 22:00	13:30 22:00	16:25 23:55	RD	41h 00m	41h 00m	5
17	SP	SP	SP	SP	SP	SP	SP	35h 00m	35h 00m	—
18	08:30 16:30	07:00 15:30	RD	RD	07:00 15:30	RD	06:20 13:45	24h 25m	32h 25m	3
19	RD	RD	07:00 15:30	07:00 15:30	07:00 15:30	07:00 15:30	06:20 14:50	42h 30m	42h 30m	5
20	RD	RD	13:30 22:00	13:30 22:00	13:30 22:00	12:30 21:00	13:30 21:00	41h 30m	41h 30m	5
21	RD	RD	14:00 22:30	13:30 22:00	14:00 22:30	14:00 22:30	14:30 22:00	41h 30m	41h 30m	5
22	14:30 23:25	RD	RD	12:30 21:00	12:30 21:00	12:30 21:00	13:30 22:00	34h 00m	42h 55m	4
23	RD	RD	06:20 14:20	06:20 14:20	06:20 14:20	06:20 14:20	RD	32h 00m	32h 00m	4
24	RD	08:30 17:00	08:30 17:00	08:30 17:00	RD	RD	RD	25h 30m	25h 30m	3
Cover	9 SP 4	13 SP 4	15 SP 4	15 SP 4	16 SP 4	16 SP 4	12 SP 4	35h 00m	38h 13m	4.35

Early turns Late turns Cover (spare) week — four duties of seven, any turn Rest day

Mon-Sat hours average 35h 00m over 24 lines, a cover week counted as a contracted week · days worked average over the 20 working lines

What was held, what moved, and what it cost

Five weeks kept to the letter; the evening filled from the two turns outside them

Kept — the shape of a week

MEASURE	TODAY (20)	PROPOSED (24)
Working weeks that are one turn (one clock time Mon–Fri)	7 of 16	12 of 20
Weeks mixing early and late turns	7	5
Rest-day breaks that are two days or more	13 of 17	18 of 24
Single rest days between duties (not a 48h break)	4	6
Days worked in a week (lines × days)	1×3, 3×4, 11×5, 1×6	1×3, 3×4, 15×5, 1×6
Cover (spare) weeks (whole weeks)	lines 1, 7, 12, 17	lines 1, 7, 12, 17
Distinct shift times (all already on today's roster)	18	19
Mon–Sat hours, average week (the contract)	35h 00m	35h 00m
Sunday duties (not contracted; half the working lines)	8 of 16 lines	9 of 20 lines

Changed — the December headcount

PEOPLE ON DUTY	TODAY	PROPOSED	WHY
Monday to Friday	12	15	the contract: 20 working lines × 35h has to be worked somewhere
Saturday	10	12	owner's figure for December, leaning late for events
Sunday	8	9	owner's figure for December
Opening at 06:20 (07:15 Sunday)	4	4	unchanged
Through to the 23:55 close, weekday	2	3	three to close
Through to the close, Saturday	2	4	four on a Saturday
Still on duty at 22:00	4	5	five are enough from 22:00

The duty table, in today's times

TURN	WK	SAT	SUN	WK	SAT	SUN
06:20–13:35 7h 15m · Early	1		1			
06:20–13:45 7h 25m · Early	1		3 +2	1 +1		
06:20–14:00 7h 40m · Early		4		4		
06:20–14:20 8h 00m · Early	2		2	1 +1		
06:20–14:50 8h 30m · Early		3		3		
07:00–15:30 8h 30m · Early			2 +2			
07:15–15:45 8h 30m · Early			3		3	
08:00–16:00 8h 00m · Early				1 +1		
08:00–16:30 8h 30m · Early	2		2			
08:30–16:30 8h 00m · Early			1	1 +1		1
08:30–17:00 8h 30m · Early			1 +1			
10:30–19:00 8h 30m · Early						1 +1
11:00–19:30 8h 30m · Late	4		4			
12:00–20:00 8h 00m · Late		4		4		
12:30–21:00 8h 30m · Late			2 +2			
13:00–21:00 8h 00m · Late			1	1 +1		-1
13:30–21:00 7h 30m · Late		1		1		
13:30–22:00 8h 30m · Late	1		4 +3	1 +1	1 +1	
14:00–22:30 8h 30m · Late	2		2			
14:30–22:00 7h 30m · Late		2		1 -1		
14:30–23:25 8h 55m · Late			3			3
14:45–23:55 9h 10m · Late		2		2		
15:15–23:55 8h 40m · Late	2		2			
16:25–23:55 7h 30m · Late			4 +4	1 +1		

Busiest weekday · Saturday · Sunday. The first three columns are what the 20-line link does now; the last three are the proposal. Green cells grew, amber shrank, a struck-through row is a time the proposal does not use.

Why the duties land where they do: Monday to Saturday totals exactly 42,000 minutes, which is 20 × 35h to the minute, so the contract is paid. But unlike a table built day-uniform, **the weekdays here are not equal** — Mon 6,225 · Tue 7,205 · Wed 7,205 · Thu 7,780 · Fri 7,690, and Saturday 5,895. Monday is the lightest day by some margin and Thursday the heaviest; that is a property of the design rather than a fault, but it is the first thing to put to the roster office, because it decides how much cover each day really has.

People on duty, hour by hour

Cover today and proposed against the measured December 2026 timetable (arrivals and departures, weighted by train length). A darker orange hour carries more of the day's traffic.

Monday to Friday window 06:20–23:55

	05	06	07	08	09	10	11	12	13	14	15	16	17	18	19	20	21	22	23
TODAY		4	4	6	6	6	7	7	8	8	8	8	6	6	6	5	5	4	2
PROPOSED		4	6	8	8	8	8	9	10	9	8	9	7	7	7	7	6	5	3
DEC 2026 TRAFFIC	5	69	107	127	122	95	85	75	84	79	87	105	140	138	107	90	73	75	66

Saturday window 06:20–23:55

	05	06	07	08	09	10	11	12	13	14	15	16	17	18	19	20	21	22	23
TODAY		4	4	4	4	4	4	5	6	9	6	6	6	6	6	5	4	2	2
PROPOSED		5	5	6	6	6	6	6	8	10	6	6	6	6	6	6	5	3	3
DEC 2026 TRAFFIC	6	24	58	75	81	92	88	77	64	68	72	84	82	67	64	79	78	62	41

Sunday window 07:15–23:25

	05	06	07	08	09	10	11	12	13	14	15	16	17	18	19	20	21	22	23
TODAY			3	4	4	4	4	4	5	8	8	5	4	4	4	4	3	3	3
PROPOSED			3	4	4	5	5	5	6	9	9	6	5	5	4	4	4	3	3
DEC 2026 TRAFFIC			15	45	65	71	74	71	61	55	64	69	78	73	75	52	71	65	50

Spare cover is not in these figures — a cover week carries no times, so the rows are a floor. Sunday: 5 December movements fall after the 23:25 finish — the standing question on whether Sunday's window moves; the window is stored per design, so this proposal can be rebuilt to either answer.

Reading it. The proposal raises the whole day rather than reshaping it, which is what the December brief asked for — the station rosters posts (ticket office, gateline, passenger assist) plus break cover, and a post needs someone on it whether the hour carries 23 trains or 5. The one hour the search was allowed to favour is Saturday's 17:00–22:00, for events. The weekday's two peaks are a duty length apart, so cover stays fairly flat between them by arithmetic, not neglect.

The checks sheet

Hard limits first — met or not. Then the ORR good-practice fatigue factors, which report what is present and never pass or fail a design.

Hard limits — a design either meets these or cannot be run

- ✓ **More than 13 consecutive days worked** — longest possible run **6** days (today: 9)
Longest possible run is 6 days, within the 13 configured here from Chiltern practice (origin: the legacy Hidden standard). A spare week is 4 duties of 7 and the roster clerk places them, so this is the worst case the link allows.
Basis: Chiltern roster policy, citation outstanding — legacy Hidden 13-in-14 standard. Configured from Chiltern practice; the policy citation is outstanding, so this is stated as the app states it.
- ✓ **At least 12 hours between duties** — **0** rests under 12h anywhere in the rotation, Saturday-into-Sunday and line-into-line included (today: 0)
The generator refuses a design it cannot repair to this; the search here never produced one.
- ✓ **The contracted week, exactly** — **35h 00m** average Mon–Sat over the 24 lines, cover weeks counted as contracted weeks
700h of duty a week across 20 working lines. Sundays (77.25h) sit on top as RDW, as they do today. Individual weeks range 24h 25m–45h 00m; only the average is the contract.

December design figures — the staffing shape agreed for the new timetable

RULE	PROPOSAL
Four on at the open, every day	× 4 weekday · 5 Saturday · 3 Sunday
Three through to the close; four on a Saturday	× 3 weekday · 3 Saturday · 3 Sunday
Five still on duty at 22:00	× 5 weekday · 3 Saturday a 22:00 finish is not "on at 22:00"
Fourteen on a Saturday, ten on a Sunday	× 12 · 9
Cover weeks, evenly spread	× 4 weeks at lines 1, 7, 12, 17 — gaps 8, 6, 5, 5 the December shape asks for five
About 4.2 days a week worked, Mon–Sat	✓ 4.35 over the 20 working lines
No :05 or :10 times except the open and close	✓ none
Late turns slightly shorter than most earlies	× not met — lates 7h30–9h10, earlies 7h15–8h30 the longest late is longer than the longest early

Two things to settle before it is frozen

Zero was the target and one is the floor, for a reason worth understanding. The factor left is FF19 — successive start times varying by more than two hours — at three instances, down from seven. **It cannot be zero in any rotation containing both early and late weeks:** going round the cycle you must cross from earlies to lates and back again, and each crossing is by definition a jump of more than two hours. Three is the fewest of the three seeds; two is the arithmetic floor for a design with this mix of turns. **What protecting the five weeks cost.** Had they been free to be rewritten, the factor count would be the same but the rotation softer: full weekends off are **4 in 24** here against 6 in Weekday Lates, because the search had five fewer weeks to arrange around. That is the price of keeping them, and it is a real one. **If they must also stay at lines 13–17,** zero is not merely hard but impossible: week 13 works Tue–Sat and week 14 Sun–Mon, so pinned adjacent they form a single 58.9-hour seven-day window that nothing else in the grid can break, and MRSF's row can never clear. Moving them apart is what took it to 51.1.

ORR fatigue factors

Good practice guidelines — Fatigue Factors, p3 (December 2021). Δ present · ✓ clear · ● standing, a property of the operation · – not applicable. Nathan assesses against this list.

CODE	FACTOR	TODAY'S LINK	PROPOSED
FF1	Night shift covering 00:00–05:00 Time of day	– –	– –
FF2	Early shift starting 05:00–07:00 Time of day	● 24	● 25
FF3	Very early shift starting before 05:00 Time of day	✓ 0	✓ 0
FF4	Very early shift (before 05:00) over 8h Duty length	– –	– –
FF5	Day shift over 12h Duty length	✓ 0	✓ 0
FF7	Early shift over 10h Duty length	✓ 0	✓ 0
FF8b	Fewer than 2 days rest after a block of early starts Recovery time	Δ 3	✓ 0
FF11	More than 13 consecutive shifts without a 48h break Recovery time	✓ 12	✓ 11
FF10	More than 4 consecutive 12h day shifts Cumulative	✓ 0	✓ 0
FF15	More than 4 consecutive early shifts in a rotating pattern Cumulative	Δ 6	✓ 4
MRSF	More than 12 consecutive day shifts Cumulative	✓ 9	✓ 6
MRSF	More than 55 hours worked in any 7-day period Cumulative	Δ 60.8	✓ 51.1
MRSF	More than 7 consecutive 8h shifts (definition to confirm) Cumulative	Δ 9	✓ 6
MRSF	More than 6 consecutive night or early shifts in a permanent pattern Cumulative	– –	– –
FF17	Backward rotating pattern (definition to confirm) Circadian	✓ 10 backward / 12 forward	✓ 14 backward / 14 forward
FF18	Rotating pattern of about a week (definition to confirm) Circadian	● 20-line rotation · typically 4h 0m a week	● 24-line rotation · typically 2h 23m a week
FF19	Successive start times varying by more than 2 hours (definition to confirm) Circadian	Δ 6	Δ 3
FF6	Night shift over 10h Night shifts	– –	– –
FF8	Less than 2 days rest after consecutive nights Night shifts	– –	– –
FF9	Less than 14h rest in any 24h for night shifts Night shifts	– –	– –
FF12	Only one day rest after night shifts Night shifts	– –	– –
FF14	More than 4 consecutive nights in a rotating pattern Night shifts	– –	– –
FF16	More than 3 consecutive night shifts over 8h Night shifts	– –	– –
FF20	First night shift Night shifts	– –	– –
FF13	Less than 12h rest in any 24h Intervals between duties	✓ 0	✓ 0

Present: 1 (today 5) · standing: 2 · FF2 fires on every 06:20 duty, so it is a property of the station's opening time rather than of any design. A cover week's four duties are counted in the worst case for every run figure. This sheet is an aid to a conversation, not a fatigue risk assessment.

How it was built, and how to load it

Reproducible: the same starting grid, seed and objective give the same rotation on any machine

How the factors were brought down

1 • Weeks 13–17 are kept to the letter. Every one of those five weeks appears in this rotation exactly as written in Weekday Lates — not a day, not a time altered. What changed is *where they sit*: 13, 14, 15, 16 and 17 are now lines 21, 3, 6, 24 and 1. That is the whole reason the factors could come down, and it is explained in the note opposite.

2 • The evening was filled from what was left. Nine duties ran 08:30–17:00 and all nine finished at once, which is why cover fell to six from 17:00 while December traffic peaks. Three of the nine are inside week 16 and were therefore untouchable; the rest are line 9's week and line 6's Friday, and both moved to **12:30–21:00** — 8h30, the same length, so the contracted week is still paid to the minute.

3 • Then the rotation was searched fatigue-first. Every factor present charged over and above its size, four restarts a seed, three seeds, with the two coverage-preserving moves only. Factors present fell from five to one, MRSF's 55-hour row from 58.9h to 51.1h, FF8b from five instances to none, and the longest run from nine days to six.

Paste it into the workspace

Links → Import, then paste the block on page 8. Line number, then Sunday to Saturday; SP is a cover week. The workspace re-runs every check on these pages from the pasted cells, so nothing here has to be taken on trust — and the two designers can edit it there like any other design.

The same rotation is also supplied as proposal-import.txt (tab-separated, pastes directly) and proposal.json (the app's own format).

Alternatives measured

DESIGN	RUN	FF PRESENT	WKENDS	ONE-TURN	STEP	WK FIT	SCORE
WL3-24-F7 · a6234195 — Weekday Lates 3 (this proposal)	6	1	4	12/20	2h 23m	46.9	—
Same Turns · d15e1b74 — the searched proposal	6	1	6	18/20	2h 33m	58.5	—
By the Book · 0f14abc — the searched proposal	6	0	6	12/20	2h 52m	41.1	—
WL-24-EXT — Weekday Lates, before any change · a52ec588	9	5	6	7/20	4h 43m	60.8	—
WL3 before the fatigue search (duty change only) · dc77e4e8	9	5	6	7/20	4h 22m	46.9	—
WL3-24- F21 — same rules, seed 21 · 9e848c92	6	1	6	10/20	2h 57m	46.9	—
WL3-24- F34 — same rules, seed 34 · 2e33804b	6	1	5	11/20	3h 17m	46.9	—
WL2-24- R21 — no protected weeks, for comparison · 33f70893	6	1	6	14/20	2h 45m	36.5	—
Workspace default · 29ec9810 (Dec 2026 table, generated)	7	5	5	0/20	4h 16m	41.1	—
Today's 20- line link (for scale)	9	5	4	7/16	4h 0m	57.7	—

How the winner was picked: rules first (rest, run, fatigue factors present, weekends off), then *Wk fit* — how evenly the weekday cover follows the December traffic curve (lower is better; today's link scores what it scores). Two candidates tied on every rule; the fit decided it, and the runner-up is named so it can be asked for. Score is the search's own feel objective (lower is better) and is not a verdict. The workspace default is the app's own December table, generated and reordered with every switch on — a good design by every panel, and not one that resembles today's.

Weekday Lates 3 · WL3-24-F7 · a6234195 · Marylebone Roster — Links designer

LOAD IT

The rotation, ready to paste

Line number, then Sunday to Saturday. SP is a cover week. Paste the whole block into Links → Import.

1	SP	SP	SP	SP	SP	SP	SP
2	13:30-22:00	13:30-22:00	RD	RD	13:30-22:00	13:30-22:00	14:45-23:55
3	14:30-23:25	14:00-22:30	RD	RD	RD	07:00-15:30	06:20-14:50
4	07:15-15:45	06:20-13:45	06:20-13:45	RD	RD	06:20-13:45	08:00-16:00
5	07:15-15:45	06:20-13:45	06:20-13:45	06:20-13:45	RD	RD	06:20-14:50
6	07:15-15:45	06:20-13:45	06:20-13:35	06:20-13:35	06:20-13:35	RD	RD
7	SP	SP	SP	SP	SP	SP	SP
8	RD	16:25-23:55	16:25-23:55	16:25-23:55	16:25-23:55	16:25-23:55	RD
9	RD	16:25-23:55	16:25-23:55	16:25-23:55	16:25-23:55	16:25-23:55	16:25-23:55
10	RD	16:25-23:55	16:25-23:55	16:25-23:55	16:25-23:55	16:25-23:55	RD
11	RD	14:00-22:30	14:00-22:30	14:00-22:30	14:00-22:30	14:00-22:30	RD
12	SP	SP	SP	SP	SP	SP	SP
13	10:30-19:00	12:30-21:00	RD	RD	06:20-14:20	06:20-13:35	06:20-14:20
14	RD	RD	08:30-16:30	08:30-16:30	13:30-22:00	13:00-21:00	14:45-23:55
15	14:30-23:25	RD	07:00-15:30	06:20-13:45	06:20-13:45	06:20-14:20	RD
16	RD	06:20-14:20	12:30-21:00	13:30-22:00	13:30-22:00	16:25-23:55	RD
17	SP	SP	SP	SP	SP	SP	SP
18	08:30-16:30	07:00-15:30	RD	RD	07:00-15:30	RD	06:20-13:45
19	RD	RD	07:00-15:30	07:00-15:30	07:00-15:30	07:00-15:30	06:20-14:50
20	RD	RD	13:30-22:00	13:30-22:00	13:30-22:00	12:30-21:00	13:30-21:00
21	RD	RD	14:00-22:30	13:30-22:00	14:00-22:30	14:00-22:30	14:30-22:00
22	14:30-23:25	RD	RD	12:30-21:00	12:30-21:00	12:30-21:00	13:30-22:00
23	RD	RD	06:20-14:20	06:20-14:20	06:20-14:20	06:20-14:20	RD
24	RD	08:30-17:00	08:30-17:00	08:30-17:00	RD	RD	RD

Prepared 17 September 2026. Every figure in this document was computed by the Marylebone Roster app's Links modules from exactly these cells; import them and the workspace's Design checks, hard limits, fatigue factors and coverage cards will restate them.